

Student Name:

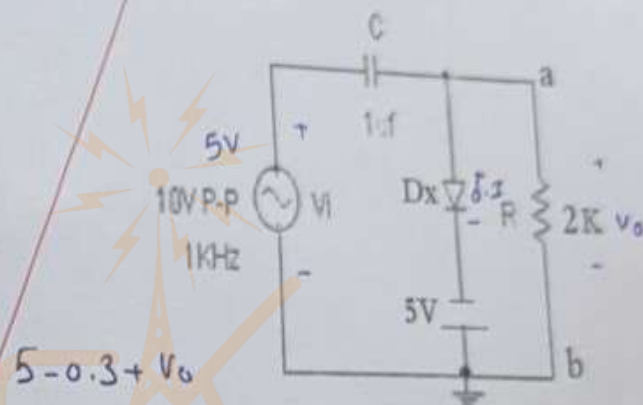
Student ID:

1. Answer the following questions.

1. If a digital multimeter (DMM) shows an (O.L) reading while testing a correctly functioning diode, this means the positive terminal of the DMM is connected to and the negative terminal is connected to to.....
 a. Anode, Cathode ☒ b. Cathode, Anode c. Can not be determined.

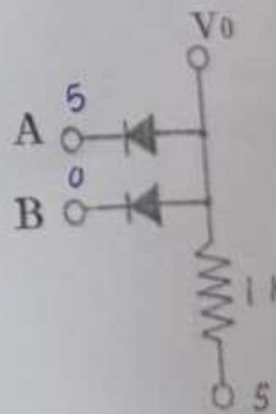
2. The circuit shown is a circuit, and the measured voltage across terminals a and b during the positive input cycle is..... (Assume the diode D_x is a simplified practical model with $V_{Knee} = 0.3 V$)

- a. Clipper, 4.7 V
 b. Clamper, -5.3 V
 c. Clipper, -4.7 V
 d. Clamper, 5.3 V
☒ e. Clamper, -4.7 V
 f. None of them.



3. The used device in the lab to display the output signal of clamper circuit is called
 a. Function generator b. DC supply c. Transformer ☒ d. Oscilloscope.

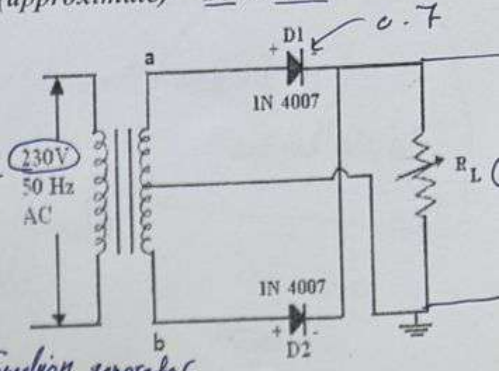
4. For the circuit below, if $A = 5 V$ and $B = 0 V$, then the output value (V_o) is
 a. 5 V ☒ b. 0 V c. Can not be determined.



the following circuit to answer the following questions where D1 and D2 are "IN 4007" (simplified practical model), $R_L = 2K\Omega$, and V_{ab} (approximate) = 14 to 16 V.

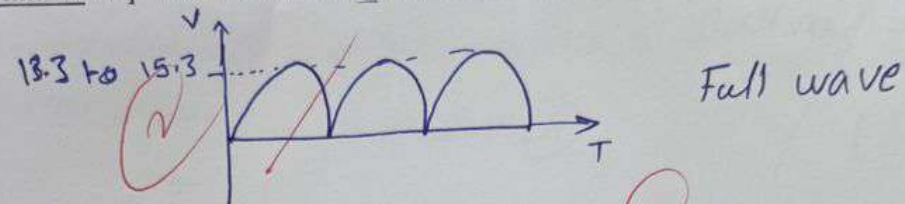
2.1. What is the purpose of this circuit?

it's rectifier (center tap) Full wave
 → it's transform from AC to DC and get Full wave
 2. Based on lab experiments, what is the device used to supply the V_{ab} voltage?



~~LM317 in DC mode~~ ~~function generator~~
 voltage

2.3. Draw an approximate output on the resistor R_L . (No need to determine values on the output)



2.4. What is the expected output frequency of the given circuit?

Full wave so $f_{out} = 2 f_{in} \Rightarrow f_{out} = 100 \text{ Hz}$

2.5. What is the effect of adding a capacitor in parallel to the resistor R_L ?
 رح ينزل الموجة انهم و خطية اكثر
 it'll be decrease the ripple

it's make the ~~output~~ output signal look more dc signal more linear

2.6. Based on Q2.5, what is the effect of decreasing the capacitor value on the effect in Q2.5?

when you decreasing the capacitor value
 the ripple will increasing and it's bad

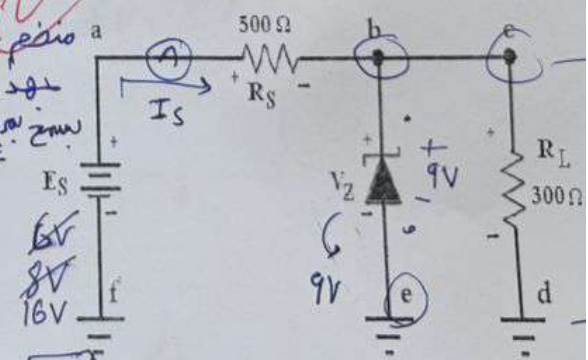
Q3. For the circuit below, the same practical model diode used in the lab is attached in the circuit. Assume the value of $V_Z = 9 \text{ V}$, answer the following.

3.1. What is the primary purpose of the given diode?

it's voltage regulator

2. If $E_S = 6 \text{ V}$, what reading will the DMM display when it is connected between terminals b and c?

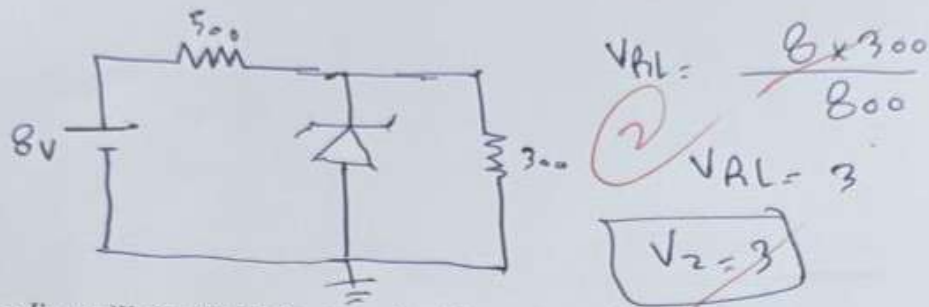
when you connected DMM between b and c



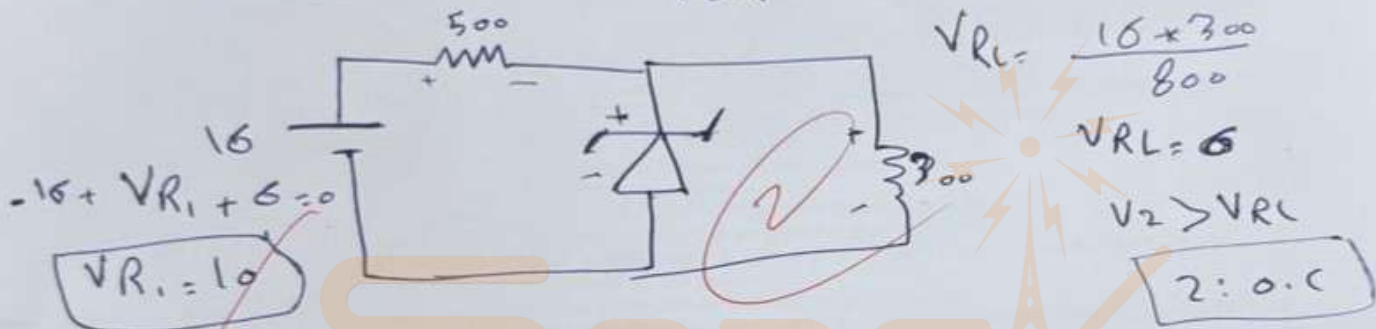
$$\frac{V_{RL}}{R_L} = \frac{7.25}{300} = 7.5 \text{ mA}$$

it's Ammeter
 → reading I_L
 Load resistance

If $E_s = 8\text{ V}$, what reading will the DMM display when it is connected on the terminals b and e?



3.4. If $E_s = 16\text{ V}$, what reading will the DMM display when it is connected on the terminals a and b?



3.5. If $E_s = 20\text{ V}$, what reading will the DMM display when measuring the current through the given diode?

